

# PARKER HUNT

Chestnut Hill, MA | huntpb@bc.edu | parkerhunt.github.io  
Department of Physics & Schiller Institute for Integrated Science and Society  
Boston College

## EDUCATION

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### Boston College | Chestnut Hill, MA

*Ph.D. Candidate in Physics* (Expected Graduation: Summer 2027)

- Research Specialization: Atmospheric Dynamics & Idealized Climate Modeling
- Advisor: Dr. Yi Ming

### The Ohio State University | Columbus, OH

*B.S. in Physics with Minor in Mathematics* (May 2020)

- Honors: *Magna Cum Laude*, with Research Distinction in Physics
- GPA: 3.85 / 4.00 (*Physics Major GPA: 3.90 / 4.00*)

## PUBLICATIONS & MANUSCRIPTS IN PREPARATION

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1. **Hunt, P.**, Lin, P., Kim, S., MacDonald, C. G., & Ming, Y. (2026). Time-mean tropical circulation in an idealized moist GCM: Dependence on Zonally Asymmetric Forcings. (*Submitted to Journal of Climate*).
2. **Hunt, P.**, Lin, P., Kim, S., MacDonald, C. G., & Ming, Y. (2026). Time-mean tropical circulation in an idealized moist GCM: Response to CO<sub>2</sub> forcings. (*In preparation*).
3. Kim, S., **Hunt, P.**, & Ming, Y. (2026). Zonal Structure of Warming-Induced Tropical Circulation Changes. (*Submitted to Geophysical Research Letters*).
4. Landry, K., Zhou, Y., Ruan, X., Chen, J.-J., Zhai, R., Kong, T., Lueders-Dumont, J., Dong, S., Osorio-Rodriguez, D., Pavia, F. J., Subhas, A. V., Adkins, J. F., Berelson, W. M., Peng, X. N., **Hunt, P.**, Tian, H., Ming, Y., & Wang, X. T. (2026). Thickening of oxygen-deficient waters in the tropical Pacific. (*In preparation*).
5. **Hunt, P.**, Lin, P., & Ming, Y. (2026). Time-mean Tropical Circulation in an Idealized Moist GCM: Divergent Circulation's Dependence on Length Scale. (*In preparation*).

## RESEARCH EXPERIENCE

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### Graduate Research Assistant | Boston College (08/2021 – Present)

*Department of Physics & Schiller Institute for Integrated Science and Society*

- Advisor: Dr. Yi Ming (08/2023 – Present) | *Geophysical Fluid Dynamics*
  - Constructed a diagnostic framework to isolate fundamental physical mechanisms which influence the tropical atmospheric circulation.
  - Evaluated the response of Hadley and Walker circulations to zonally asymmetric forcings and elevated CO<sub>2</sub> concentrations in an idealized general circulation model.
  - Contributed to methodology used to investigate the tropical circulation response to warming scenarios in CMIP6 model ensemble.
  - Evaluated the time-mean surface wind trends in ERA5 reanalysis product and their connection to the divergent circulations.
  - Configured and executed idealized climate simulations on high-performance computing (HPC) clusters utilizing Fortran and Python within a Unix/Linux environment.
  - Mentored undergraduate researchers in computational workflows and high-performance computing techniques.
- Advisor: Dr. Benedetta Flebus (08/2021 – 08/2023) | *Theoretical Condensed Matter Physics*
  - Evaluated Magnon-phonon interactions in non-Hermitian second-quantized systems.

### Undergraduate Research Assistant | The Ohio State University (10/2016 – 05/2021)

*Department of Physics*

- Advisor: Dr. Ciriya Jayaprakash (04/2019 – 05/2021) | *Non-Linear Dynamical Systems*
  - Developed 2D discretized numerical simulations and 1D continuous partial differential equations (PDEs) in C++ and Mathematica to examine interface motion and domain stability of spatially extended stochastic dynamical systems.
  - Investigated accessible stable states available to the system using methods from statistical mechanics.
- Advisors: Dr. Andrew Heckler & Dr. Gregory Kilcup (04/2017 – 04/2019) | *Physics Education Research*
  - Created targeted diagnostic problem sets designed to reduce cognitive load and enhance foundational problem-solving skills in intermediate mechanics.
  - Analyzed statistical data on student estimation of physical quantities to identify correlations between physical intuition and overall academic performance.
- Advisor: Dr. Jay Gupta (10/2016 – 04/2017) | *Experimental Condensed Matter Physics*
  - Worked to construct a tabletop Raman spectrometer to conduct optical imaging and quality characterization of 2D materials and monolayer graphene.
  - Utilized Zeiss optical microscopy to evaluate micro-structural properties of synthesized graphene samples.

## TALKS AND POSTERS

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### Conference Presentations

- **Hunt, P.**, Kim, S., MacDonald, C. G., & Ming, Y. (2025). *Convective reorganization under different zonally-asymmetric thermal forcings and CO<sub>2</sub> concentrations in an idealized model*. AGU Fall Meeting, New Orleans, LA.
- **Hunt, P.**, Kim, S., & Ming, Y. (2025). *Hadley circulation under different zonally-asymmetric thermal forcings and CO<sub>2</sub> concentrations in an idealized model*. Gordon Research Conference (GRC), Lewiston, ME.
- **Hunt, P.**, Kim, S., & Ming, Y. (2024). *Interconnectivity of tropical circulation patterns*. ERDEC Conference, Chestnut Hill, MA.

### Departmental Seminars

- **Hunt, P.**, Pu, L., Kim, S., MacDonald, C. G., & Ming, Y. (2026). *Convective Reorganization*. Department of Earth and Environmental Sciences Seminar, Boston College.
- **Hunt, P.** (2026). *Topological Origins of Equatorial Waves*. Department of Physics Seminar, Boston College.

## TEACHING EXPERIENCE

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### Graduate Teaching Assistant | Boston College (08/2021 – Present)

- Instructed lab and recitation sections across many distinct physics courses, including: Graduate Electrodynamics, Introductory Physics (Mechanics and Electromagnetism), Classical Mechanics for Engineers, Physics for Pre-Medical Students, and Physics of the Environment.
- Communicated complex physical principles to diverse student cohorts, led interactive problem-solving sessions, and provided detailed feedback on coursework.

### Undergraduate Teaching Assistant | The Ohio State University (08/2018 – 05/2021)

- Facilitated flipped-classroom discussion sections for 200+ students in Intermediate Mechanics (PHYS 2300/2301) utilizing Morin's *Classical Mechanics* textbook.
- Conducted midterm review workshops for 40+ students and held individual tutoring sessions to reinforce theoretical concepts.
- Led lab and recitation sections for introductory calculus-based mechanics, adapting curriculum delivery to remote learning formats via Zoom.

## TECHNICAL & COMPUTATIONAL SKILLS

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- Programming Languages: Python, Fortran, C++, Mathematica, Java
- Computational & Modeling Methods: High-Performance Computing (HPC / Linux / Unix), Idealized Climate Modeling, PDE Numerical Simulations, Stochastic Systems

## AWARDS & HONORS

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- 3rd Place, NeurIPS Competition on Reinforcement Learning, Boston University (2022)
- James Smith Scholarship, The Ohio State University (2019)

- Undergraduate Summer Research Scholarship, The Ohio State University (2018)
- Dean's List (7 of 8 Semesters), The Ohio State University
- Freshman Physics Academic Achievement Award, The Ohio State University

## ACADEMIC OUTREACH

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**Student Board Member** | Schiller Institute for Integrated Science and Society, Boston College (2024 – Present)

- Planned and hosted informational seminars on research opportunities and student involvement across disciplines.
- Founded a service sub-committee dedicated to community service initiatives in the greater Boston area.
- Served as the graduate student representative during interviews and evaluation processes for new faculty candidates.

**Event Volunteer** | Cloud Feedback Model Intercomparison Project (CFMIP) Meeting (2024)

- Assisted with event coordination, technical setup, attendee check-in, and poster session management for international climate research attendees.

**Event Volunteer** | APS Conference for Undergraduate Women in Physics (CUWiP) (2023)

- Facilitated conference operations, handled attendee registration, managed room assignments, and directed poster session setups.

## RELEVANT GRADUATE COURSEWORK

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- Geophysical Modeling (Dept. of Earth & Environmental Sciences, BC)
- Mathematical Physics (Dept. of Physics, BC)
- Climate, Energy, and Sustainability Nexus (Schiller Institute, BC)
- Artificial Intelligence and Deep Learning (Dept. of Computer Science, BU)
- Big Data Analytics & AI Methods (Dept. of Physics, OSU)
- Particle Physics (Dept. of Physics, BC)
- Topology in Condensed Matter (Dept. of Physics, BC)
- Computational Physics (Dept. of Physics, OSU)
- Applied Differential Geometry in General Relativity (Dept. of Mathematics, OSU)

## REFERENCES

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### **Dr. Yi Ming**

Professor, Department of Earth and Environmental Sciences  
Core Faculty, Schiller Institute for Integrated Science and Society  
Boston College  
Relationship: Ph.D. Dissertation Advisor  
Email: [Ming.Yi@bc.edu](mailto:Ming.Yi@bc.edu)

### **Dr. Pu Lin**

Senior Research Scientist, Department of Earth and Environmental Sciences  
Boston College  
Relationship: Primary research collaborator & Co-Author  
Email: [Lin.Pu@bc.edu](mailto:Lin.Pu@bc.edu)

### **Dr. Cameron MacDonald**

Environmental Fellow (Postdoctoral Researcher)  
Harvard University  
Relationship: Research collaborator & Co-Author  
Email: [cgmacdona1d@fas.harvard.edu](mailto:cgmacdona1d@fas.harvard.edu)

### **Dr. Ciriya1m Jayap1rakash**

Professor Emeritus, Department of Physics  
The Ohio State University  
Relationship: Undergraduate thesis advisor  
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